

State of Developer Ecosystem 2024: Language Trends, Satisfaction

Key insights from JetBrains' survey of 23,262 developers on language usage, satisfaction, adoption intent, and domain specialization

Audience: engineering leaders, CTOs, and developer experience teams

Narrative: how usage is changing → what developers prefer → what to do about it

Scope: language trends, satisfaction, adoption intent, and domain specialization

Python Closes the Gap: 8 Years of Language Usage Shifts

Language	2017	2020	2022	2024
JavaScript	65	70	65	61
Python	32	55	53	57
HTML/CSS	60	61	54	51
SQL	42	56	49	48
Java	47	54	48	46
TypeScript	12	28	34	37
Go	8	19	19	18
Rust		7	9	11
PHP	30	27	20	17
Kotlin	2	17	16	14

JavaScript: 70% peak (2020) → 61% (2024)

Python: 32% (2017) → 57% (2024)

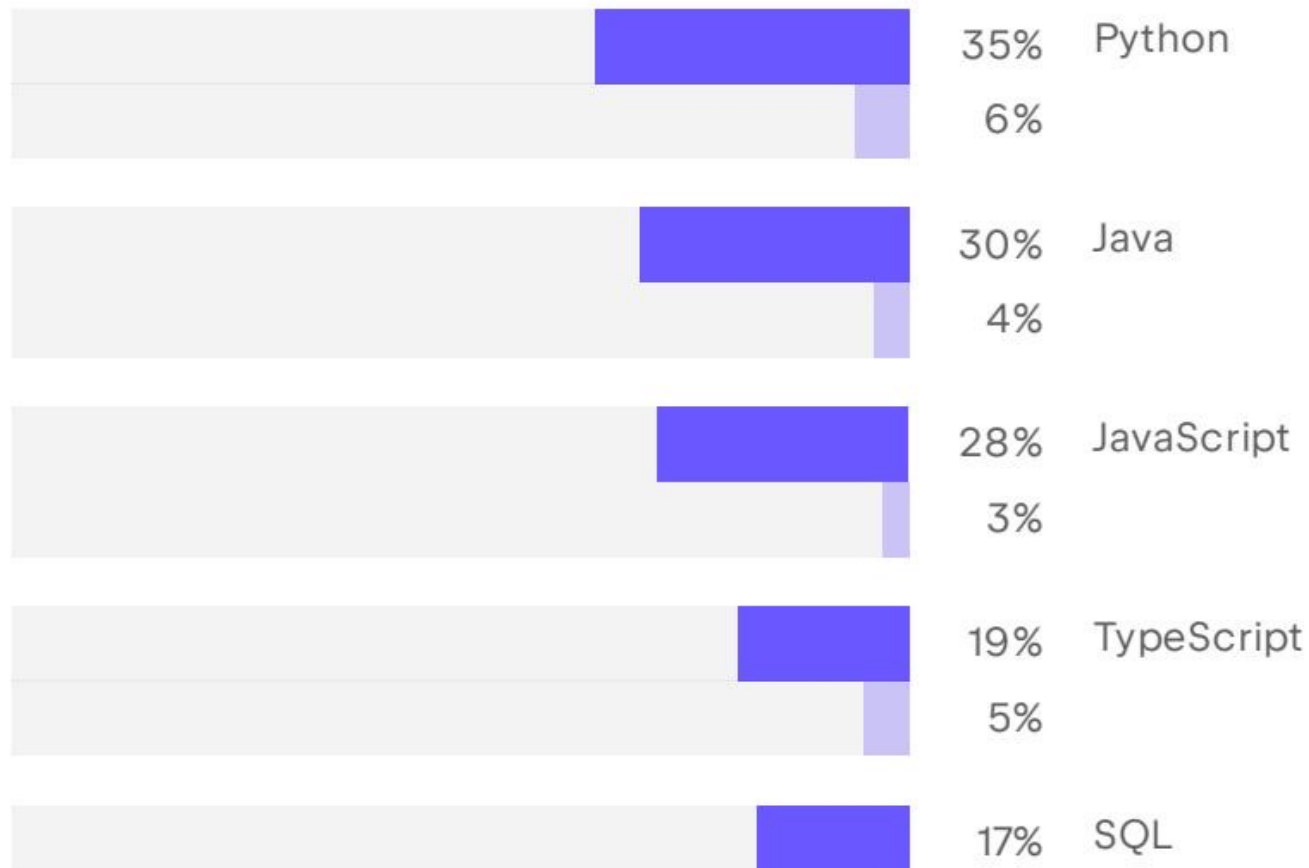
TypeScript: 12% (2017) → 37% (2024); PHP: 30% → 17%

Python Is Now the #1 Primary Language at 35%

Primary language (share of developers)

■ Primary language

■ Planning to adopt



Top primary languages

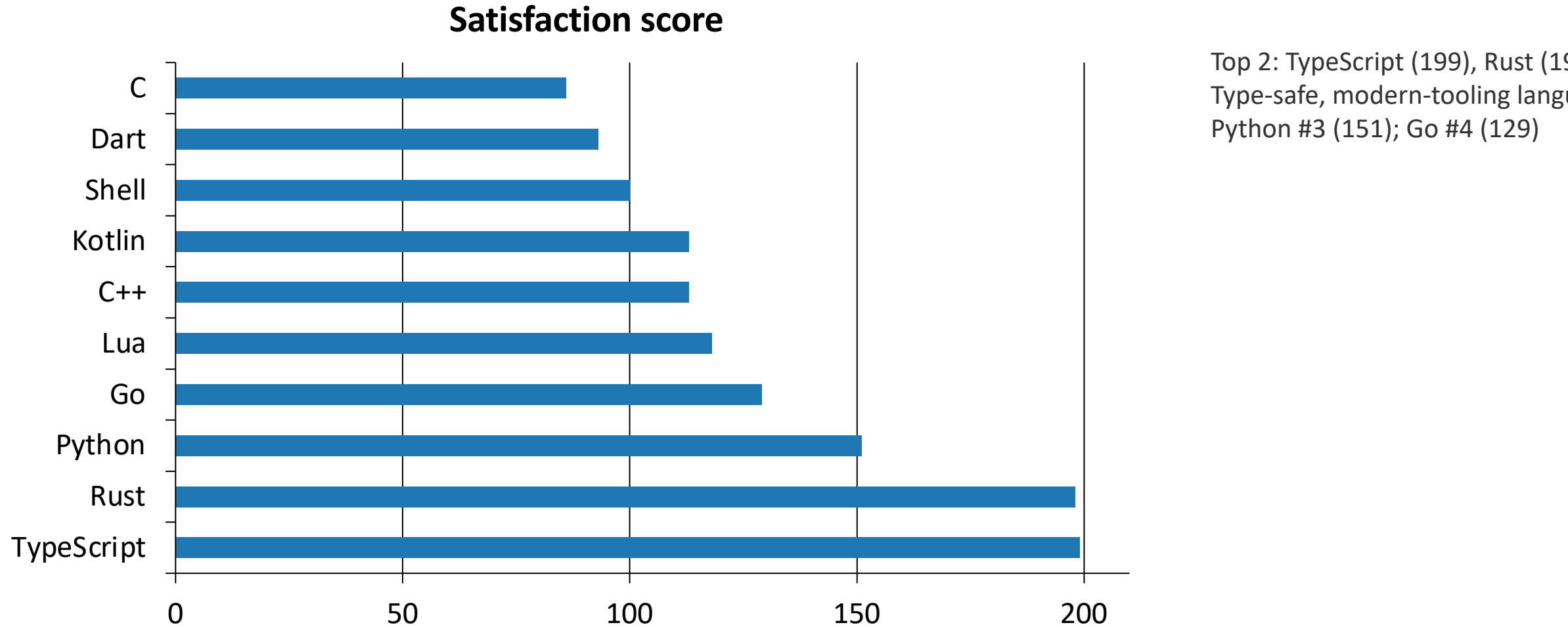
Python: 35%
Java: 30%
JavaScript: 28%
TypeScript: 19%
SQL: 17%

Adoption intent stays low

Python: 6%
TypeScript: 5%
Java: 4%
SQL: 4%

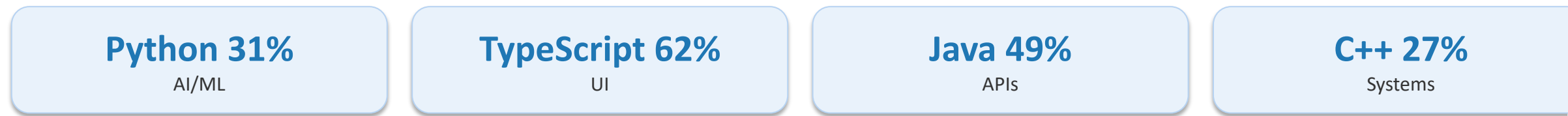
Ecosystem lock-in: few plan to switch primary stacks (<6%)

TypeScript and Rust Top Developer Satisfaction, Nearly Tied at



Each Language Owns a Domain: AI Belongs to Python, UI to TypeScript

Application domain	Top language	Share
AI / Machine Learning	Python	31%
Data Science / Data Analytics	Python	28%
Data Processing & Analytics	Python	17%
User Interface	TypeScript	62%
APIs & Services	Java	49%
Application Logic & Workflows	Java	55%
System Tools & Components	C++	27%
Integrating with APIs & Services	Python	38%



Source: JetBrains language use-cases by domain (2024)

What This Means for Your Technology Strategy in 2025

Invest in Python for AI/ML teams — it is the undisputed leader
TypeScript is the future of frontend and increasingly full-stack
Rust and Go are the growth plays for infrastructure and systems

PHP (and Ruby) long-term hiring pools are shrinking
Satisfaction correlates with type safety + modern tooling
Prioritize domain-fit to guide stack and hiring decisions

Bottom line: align language bets to domains, and optimize developer experience to retain talent.